Channels

SWAT+ Documentation

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Purpose

The `channel` line of `file.cio` names eight channel input files. The table below gives the use of each in 62.0.1. `initial.cha` (initial channel contents), `temperature.cha` (stream temperature) and the legacy files `channel.cha`, `hydrology.cha` and `sediment.cha` are described below; the other files are on the chandeg channel page.

SWAT+ has two channel objects. The channels listed in `chandeg.con` (SWAT-DEG or "lte" channels) are the ones 62.0.1 routes; their parameter files are on the chandeg channel page. The legacy channel object listed in `channel.con` is set up at the start of a run, but its daily routing is switched off in 62.0.1, so water sent to it does not leave it. Build channel networks with `chandeg.con`.

Files on the channel line

Each slot can be `null`. A missing file is treated like `null`.

Slot	Default name	Used in 62.0.1 for	Documented on
1	`initial.cha`	Initial water, sediment, nutrients and pesticides in chandeg channels.	initial.cha
2	`channel.cha`	Legacy channel object only. No effect on results.	legacy files

Slot	Default name	Used in 62.0.1 for	Documented on
3	`hydrology.cha`	Legacy channel object only. No effect on results.	legacy files
4	`sediment.cha`	Legacy channel object only. No effect on results.	legacy files
5	`nutrients.cha`	In-stream water quality of chandeg channels.	chandeg channels
6	`channel-lte.cha`	One row per chandeg channel, pointing to the other files.	chandeg channels
7	`hyd-sed-lte.cha`	Geometry, hydraulics and sediment properties of chandeg channels.	chandeg channels
8	`temperature.cha`	Read, but the stream temperature calculation uses only built-in values.	temperature.cha

Three more channel files have fixed names and are read when present in the run folder: `initial.cha_cs` (initial pathogens, salts and constituents, below), `sed_nut.cha` (chandeg channels) and `shade_factor.shf` (below).

initial.cha

Named sets of initial conditions for chandeg channels. Each row of `channel-lte.cha` selects a set by name in its `cha_ini` column. Free format: two header lines are skipped, then one set per line. There is no `id` column, so the first value on each line is the name. The Editor adds a trailing `description` value, which the reader ignores.

#	Column	Description
1	`name`	Name of the set, matched against `cha_ini` in `channel-lte.cha`.
2	`org_min`	Name of a row in `om_water.ini`: initial water, sediment and nutrient concentrations.
3	`pest`	Name of a row in `pest_water.ini`: initial pesticide in the water and the bed.
4	`path`	Read but not used. Initial pathogens come from `initial.cha_cs`.
5	`hmet`	Read but not used.
6	`salt`	Read but not used. Initial salts come from `initial.cha_cs`.

A `cha_ini` name not found in `initial.cha`, or an `org_min` or `pest` name not found in its file, falls back to row 1 of that file without a message. Keep `initial.cha` and an `om_water.ini` in any model with chandeg channels.

Initial water and constituents

The `flo` value of the `om_water.ini` row is the initial water depth as a fraction of the bankfull depth `dp`. The channel starts with the volume the rating curve gives at that depth, and the concentrations of the row are converted to masses with that volume. With `flo` of 1 or less all of it is channel storage. Above 1 the volume is split between channel and floodplain in proportion to their volumes on the rating curve. Channels 0.001 km (1 m) long or shorter start empty.

Initial pesticides in the water are the `pest_water.ini` concentration (mg/kg) times the stored water; pesticides in the bed are the benthic concentration times the volume of the active bed layer (width times length times the pesticide's active bed depth).

initial.cha_cs

An optional file with a fixed name, read when present, with initial pathogens, salts and other constituents for chandeg channels. Two header lines are skipped, then one row per line:

#	Column	Description
1	`name`	Name of the row. Not checked.
2	`pest`	Read but not used. Pesticides come from `initial.cha`.

#	Column	Description
3	`path`	Name of a row in `path_water.ini`.
4	`hmet`	Read but not used.
5	`salt`	Name of a row in `salt_channel.ini`: initial salt ion concentrations (g/m^3).
6	`cs`	Name of a row in `cs_channel.ini`: initial constituent concentrations (g/m^3).

Rows are paired with `initial.cha` by position: row 3 of `initial.cha_cs` applies to channels that use row 3 of `initial.cha`, whatever its name. Keep the two files in the same order with the same number of rows. Salt and constituent masses are the concentrations times the initial water volume.

In 62.0.1 the pathogen pointer is resolved, but the initial pathogen values are then read from the `path_water.ini` row with the position of the channel's `pest_water.ini` row.

temperature.cha

Parameters of the stream temperature calculation for chandeg channels. Free format: two header lines are skipped, then one set per line with 15 values. The name is limited to 13 characters.

#	Column	Units	Built-in value	Typical range	Description
1	`name`	-	-	-	Name of the set (up to 13 characters).
2	`sno_mlt`	-	1.0	-	Weight of the snowmelt contribution.
3	`gw`	-	1.0	-	Weight of the groundwater contribution.
4	`sur_lat`	-	1.0	-	Weight of the surface runoff and lateral flow contributions.
5	`sno_lag`	days	2	1 to 3	Air temperature averaging period for snowmelt.
6	`gw_lag`	days	250	200 to 365	Air temperature averaging period for groundwater.
7	`surf_lag`	days	5	2 to 5	Air temperature averaging period for surface runoff.
8	`lat_lag`	days	10	5 to 10	Air temperature averaging period for lateral flow.
9	`lat_lag_coef`	-	0.75	-	Multiplier on the lagged air temperature for lateral flow.
10	`surf_lag_coef`	-	0.75	-	Multiplier on the lagged air temperature for surface runoff and snowmelt.
11	`gw_lag_coef`	-	0.75	-	Multiplier on the lagged air temperature for groundwater.
12	`hex_coef1`	-	0.75	-	Multiplier on the dew point temperature in the heat exchange.
13	`hex_coef2`	-	1.5	-	Divisor on the channel depth when the flow depth for heat exchange is calculated.
14	`sf_on`	-	0	0 or 1	Shade factor source: `1` daily values from `shade_factor.shf`, `0` the single value `ssff`. Integer.
15	`ssff`	fraction	0.5	0 to 1	Shade factor: the fraction of solar radiation kept off the water surface.

In 62.0.1 the rows of `temperature.cha` have no effect. The reader stores them, but the temperature calculation always uses the built-in values in the table, whether or not the file exists, for every channel. With the built-in `sf_on` of 0, `shade_factor.shf` is not used either. The stream temperature calculation itself runs every day for every chandeg channel.

Stream temperature

The temperature of the water leaving a channel is calculated in three steps. Values in this section are the built-in values above.

Local inflow. For the routing units that drain directly into the channel, the day's HRU snowmelt Q_{sno}, surface runoff Q_{sur}, lateral flow Q_{lat} and water yield Q_{wyld} are summed (mm), and the groundwater inflow Q_{gw} (m³/s, at most 10) is taken from the channel's hydrograph separation. Each component gets an air temperature averaged over its lag period: T_x is the mean of the daily $(T_{max} + T_{min})/2$ over the last `surf_lag`, `lat_lag`, `gw_lag` or `sno_lag` days, at least 0.01 °C, and for snowmelt at most 2.5 °C. In the first simulation year the lag is limited to the day of the year. Then

$$T_{loc} = \frac{a_{sno} k_{sur} T_{sno} Q_{sno} + a_{gw} k_{gw} T_{gw} Q_{gw} + a_{sl} k_{lat} T_{lat} Q_{lat} + a_{sl} k_{sur} T_{sur} Q_{sur}}{Q_{wyld}}$$

with a_{sno} = `sno_mlt`, a_{gw} = `gw`, a_{sl} = `sur_lat`, and k_{sur}, k_{lat} and k_{gw} the lag coefficients `surf_lag_coef`, `lat_lag_coef` and `gw_lag_coef`. With gwflow active the groundwater term is left out. If T_{loc} differs from the previous day's outflow temperature by more than 5 °C, or the weather station has no temperature gauge, $T_{loc} = 5 + 0.75\,\bar{T}$ with $\bar{T}$ the day's mean air temperature.

Mixing with upstream water. Water from upstream channels arrives at their flow-weighted outflow temperature T_{up} (or $5 + 0.75\,\bar{T}$ for a channel without upstream channels). It is weighted by q, the outflow of the channel itself for the day (m³/s):

$$T_0 = \frac{T_{up}\, q + T_{loc}\, Q_{wyld}}{q + Q_{wyld}}$$

or $T_0 = T_{loc}$ when the outflow is zero.

Heat exchange. Following Edinger et al. (1974), with dew point $T_d = \min(h_1\, T_{dew},\ \bar{T})$, h_1 = `hex_coef1`, $T_m = (T_0 + T_d)/2$ and wind speed U (m/s),

$$\beta = 0.35 + 0.015\, T_m + 0.0012\, T_m^2, \qquad K_e = 4.48 + 0.05\, T_0 + (\beta + 0.47)(9.2 + 0.46\, U^2)$$

The short-wave input is $H_s = 0.97 \times 11.57\, R_s\, (1 - SF)$ W/m², with R_s the solar radiation (MJ/m²/day) and SF the shade factor `ssff`. The long-wave input is $H_a = 0.96\, \varepsilon\, \sigma\, (\bar{T} + 273.15)^4$ with $\varepsilon = 0.74 + 0.0065\, e_a\, (1 + 0.17\, C^2)$, e_a the vapour pressure (hPa) and $C = 0.9\, R_s / R_{s,max} + 0.1$ the cloud factor. The equilibrium temperature is

$$T_{eq} = T_d + \frac{H_s + H_a - 305.5 - 4.48\, T_d}{K_e}$$

and the outflow temperature is

$$T = T_0 + \frac{K_e\, (T_{eq} - T_0)\, t}{\rho\, c_p\, d_f / 86400}, \qquad T \geq 0.001$$

where t is the travel time through the channel (days) at the day's peak flow, $\rho c_p = 4.186 \times 10^6$ J/m³/°C and d_f the flow depth (m). The flow depth comes from a triangular section with depth-to-width ratio $(D/h_2)/W$, h_2 = `hex_coef2`, that passes the day's outflow at the channel velocity; it is at least 0.1 m and at most $2D$.

The result is the outflow temperature passed downstream and reported in the outflow block of the `channel_sd` output (see channel outputs). The in-stream water quality and the `water_temp` column of `channel_sd` use $5 + 0.75\,\bar{T}$ instead.

Shade factor file

`shade_factor.shf` has a fixed name. Two header lines are skipped, then one record per line with three values: day of year, channel number (the chandeg channel's sequence number) and shade factor (fraction). It would be used with `sf_on` = 1 on day 1 to 365 (day 366 uses 0.5); in 62.0.1 `sf_on` is always the built-in 0.

Legacy channel files

`channel.cha`, `hydrology.cha` and `sediment.cha` describe the legacy channel object of `channel.con`. SWAT+ 62.0.1 reads them and sets up those channels at the start of the run, but does not route water through them, so the values have no effect on results. The formats are given for existing models. Two header lines are skipped in each file and the reader does not match columns by header name; the column names below are the names the reader uses.

channel.cha

#	Column	Description
1	`id`	Channel number.
2	`name`	Channel name.
3	`init`	Name of a row in `initial.cha`.
4	`hyd`	Name of a row in `hydrology.cha`.
5	`sed`	Name of a row in `sediment.cha`.
6	`nut`	Name of a row in `nutrients.cha`.

Names that are not found are written to `diagnostics.out` as `<name> not found (hydrology.cha)` and similar.

hydrology.cha

#	Column	Units	Built-in value	Description
1	`name`	-	-	Name of the set.
2	`w`	m	2.0	Average width of the main channel.
3	`d`	m	0.5	Average depth of the main channel.
4	`s`	m/m	0.01	Average slope. Values of 0 or less become 0.0001.
5	`l`	km	0.1	Length. Values of 0 or less become 0.001.
6	`n`	-	0.05	Manning's n, limited to 0.01 to 0.70.
7	`k`	mm/h	0.01	Effective hydraulic conductivity of the channel alluvium.
8	`wdr`	m/m	6.0	Width-to-depth ratio. Values of 0 or less become 3.5.
9	`alpha_bnk`	days	0.03	Bank storage recession factor, stored as $e^{-\alpha}$.
10	`side`	-	0.0	Side slope (horizontal per unit rise). Values of 10^{-6} or less become 2.0.

sediment.cha

#	Column	Units	Built-in value	Description
1	`name`	-	-	Name of the set.
2	`eqn`	-	0	Sediment routing method: `0` original SWAT, `1` Bagnold, `2` Kodatie, `3` Molinas and Wu, `4` Yang.

#	Column	Units	Built-in value	Description
3	cov1	-	0.1	Channel erodibility factor. Limited to 0 to 1 for eqn 0; for the other methods, values of 0 or less become 1 and values above 25 become 25.
4	cov2	-	0.1	Channel cover factor, with the same limits as cov1.
5	bnk_bd	g/cm³	0	Bank bulk density. Values near 0 become 1.40.
6	bed_bd	g/cm³	0	Bed bulk density. Values near 0 become 1.50.
7	bnk_kd	cm³/N/s	0	Bank erodibility from a jet test. Values near 0 become $0.2/\sqrt{\tau_c}$, or 0.2 without a critical shear stress.
8	bed_kd	cm³/N/s	0	Bed erodibility, as bnk_kd.
9	bnk_d50	µm	0	Bank median particle size. Values near 0 become 50; at most 10000.
10	bed_d50	µm	0	Bed median particle size. Values near 0 become 500; at most 10000.
11	tc_bnk	N/m²	0	Critical shear stress of the bank.
12	tc_bed	N/m²	0	Critical shear stress of the bed.
13 to 24	erod(1) to erod(12)	-	0	Monthly erodibility, 0 non-erosive to 1 no resistance. If all twelve are zero, all are set to cov1.

Example

refdata/Osu_1hru in the SWAT+ repository, file.cio channel line and initial.cha:

```
channel            initial.cha       null              null              null              nutrients.cha

initial.cha: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                        org_min             pest             path             hmet               salt
initcha1                    no_init             null             null             null               null
```

The model uses only chandeg channels: slots 2 to 4 and 8 are null. All channels use the initial set initcha1, whose org_min points to the row no_init of om_water.ini; that row has all values zero, so the channels start empty. The pest, path, hmet and salt pointers are null. No reference dataset includes temperature.cha, initial.cha_cs, channel.cha, hydrology.cha or sediment.cha.

Related files

- Chandeg channel files: channel-lte.cha, hyd-sed-lte.cha, sed_nut.cha and nutrients.cha.
- chandeg.con and channel.con: the two channel object types.
- file.cio: the channel line.
- om_water.ini, pest_water.ini, path_water.ini, salt_channel.ini, cs_channel.ini: initial contents referenced from initial.cha and initial.cha_cs.
- tmp.cli: the daily maximum and minimum temperatures used for the lagged air temperatures.
- Channel outputs and channels in the model theory.

Sources

- File name slots: type input_cha in input_file_module.f90, read in readcio_read.f90; in_shf in the same module
- Read order: proc_cha.f90; temperature.cha and shade_factor.shf from proc_read.f90
- Readers: ch_read_init.f90, ch_read_init_cs.f90, ch_read_temp.f90, shade_factor_read.f90, ch_read.f90, ch_read_hyd.f90, ch_read_sed.f90
- Type definitions: channel_data_module.f90 (channel_init_datafiles, channel_init_datafiles_cs, water_temperature_data, channel_data_char_input, channel_hyd_data, channel_sed_data); swatdeg_init_datafiles in sd_channel_module.f90; shade_factor_data in hydrograph_module.f90

- Initial pointers and contents: sd_channel_read.f90, sd_hydsed_init.f90, om_water_init.f90
- Stream temperature: ch_temp.f90, called from sd_channel_control3.f90; travel time and velocity from sd_channel_sediment3.f90
- Legacy object: set-up in ch_initial.f90; the "chan" case is commented out in command.f90

File and column	Fortran variable
initial.cha row	ch_init(i) (name, org_min, pest, path, hmet, salt)
initial.cha_cs row	ch_init_cs(i) (name, pest, path, hmet, salt, cs)
resolved pointers	sd_init(i)%org_min, %pest, %path, %salt, %cs
temperature.cha row	w_temp(i), components as the column names
shade_factor.shf row	shf_db(i)%jday, %cha, %value
channel.cha row	ch_dat_c(i) -> ch_dat(i)
hydrology.cha row	ch_hyd(i)
sediment.cha row	ch_sed(i)

Purpose

SWAT+ 62.0.1 routes water, sediment, nutrients and other constituents through the channels listed in chandeg.con. These are often called SWAT-DEG or "lte" channels, and their outputs are the channel_sd files. They use these parameter files:

- channel-lte.cha: one row per channel, pointing by name to the other files.
- hyd-sed-lte.cha: channel geometry, hydraulics, bank and bed properties.
- sed_nut.cha: optional extra sediment and nutrient coefficients. The SWAT+ Editor does not write it.
- nutrients.cha: in-stream water quality (QUAL2E) rate constants.

Initial channel contents come from initial.cha, described on the channel group page together with temperature.cha and the legacy files channel.cha, hydrology.cha and sediment.cha. The legacy channel object of channel.con is not routed in 62.0.1, so those legacy files have no effect on the results.

Links between the files

The lcha column of chandeg.con gives the id of a row in channel-lte.cha. That row names one row in each of the other files:

channel-lte.cha column	Matched against	If the name is not found
cha_ini	name in initial.cha	row 1 of initial.cha
cha_hyd	name in hyd-sed-lte.cha, and name in sed_nut.cha	row 1 of each file (or the built-in values of sed_nut.cha when that file is absent)
cha_sed	not used	
cha_nut	name in nutrients.cha	row 1 of nutrients.cha

An unmatched name writes no message to diagnostics.out; the channel takes row 1 without warning. Check the names before a run.

The file names of channel-lte.cha, hyd-sed-lte.cha, nutrients.cha and initial.cha are set on the channel line of file.cio (slots 6, 7, 5 and 1). sed_nut.cha has a fixed name and is read when a file of that name is in the run folder.

channel-lte.cha

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one channel with six values in this order.

#	Column	Units	Description
1	id	-	Channel number. chandeg.con refers to the row by this number in its lcha column.
2	name	-	Channel name.
3	cha_ini	-	Name of a row in initial.cha.
4	cha_hyd	-	Name of a row in hyd-sed-lte.cha, and of a row in sed_nut.cha when that file is used.
5	cha_sed	-	Read but not used. The Editor writes null.
6	cha_nut	-	Name of a row in nutrients.cha.

Number the rows 1, 2, 3 and so on in file order. The row is stored under its id, but the name lookups are applied to the row's position in the file, so rows whose id differs from their position get the wrong parameter sets.

hyd-sed-lte.cha

Free format. Two header lines are skipped, then one parameter set per line with 23 values in this order. The reader does not match columns by header name. The Editor adds a description header with no values under it, which the reader ignores.

Several column headers written by the Editor do not describe what SWAT+ 62.0.1 does with the value. The table gives the Editor header and the use in 62.0.1.

#	Column	Units	Typical range	Description
1	name	-	-	Name of the parameter set, matched against cha_hyd in channel-lte.cha.
2	order	-	1 to 12	Stream order (integer). Groups channels in the chanbud_order output and is tested by the decision table condition ch_order. Use 1 to 12: values outside that range fall outside the order budget table.
3	wd	m	from QSWAT+	Bankfull top width of the channel, calculated by QSWAT+ from the drainage area.
4	dp	m	from QSWAT+	Bankfull depth, from the top of the bank to the channel bottom, calculated by QSWAT+ from the drainage area.
5	slp	m/m	from QSWAT+	Average channel slope along its length, from the DEM. Values below 10^{-9} are raised to 10^{-6}.
6	len	km	from QSWAT+	Channel length. A channel of 0.001 km (1 m) or less passes its inflow through unchanged.
7	mann	-	0.025 to 0.15	Manning's n of the channel. Values below 0.01 are raised to 0.01. Chow (1959) gives about 0.03 for clean straight channels, 0.04 to 0.05 for winding channels with pools, weeds or stones, and 0.07 to 0.15 for sluggish or very weedy reaches.

#	Column	Units	Typical range	Description
8	`k`	mm/h	0 to 130	Effective hydraulic conductivity of the channel bed, used for transmission losses. Lane (1983) gives more than 127 mm/h for very clean gravel and large sand, 51 to 127 for clean sand and gravel, 25 to 76 for sand and gravel with little silt and clay, 6 to 25 for sand and gravel with much silt and clay, and 0.025 to 2.5 for consolidated beds with high silt and clay. Use 0 for perennial streams fed by groundwater.
9	`erod_fact`	-	-	Bank erosion exponent. Read but not used.
10	`cov_fact`	-	0 to 1	Channel cover factor. It adds a vegetation term to the critical velocity for bank erosion, so a larger value protects the banks more. See Bank erosion.
11	`wd_rto`	-	1.05 and up	Read as the channel sinuosity (channel length over straight-line length), although the header and the values the Editor writes are a width-to-depth ratio. Values below 1.05 are raised to 1.05. Used for the bend velocity in bank erosion.
12	`eq_slp`	-	about 1	Critical velocity coefficient: a multiplier on the calculated critical velocity for bank erosion. The header name suggests an equilibrium slope; 62.0.1 has no such parameter.
13	`d50`	mm	-	Median bed sediment size. Read, but replaced by 1.0 mm at the start of every day's sediment calculation, so the value in the file has no effect.
14	`clay`	%	0 to 100	Clay content of bank and bed. Sets the bank cohesion.
15	`carbon`	%	0 to 100	Organic carbon content of the bed. Sets the pesticide sorption coefficient in the channel.
16	`dry_bd`	t/m³	1.1 to 1.9	Dry bulk density of bank and bed. Converts bank and bed erosion from metres to tonnes, and sets the bed porosity for pesticide mixing.
17	`side_slp`	-	-	Channel side slope. See Channel geometry and rating curve.
18	`bed_load`	-	-	Read as the bankfull flow multiplier, the ratio of the flow at which overbank flooding starts to the bankfull flow. Replaced by 1.5 at the start of every day's sediment calculation, so the value in the file has no effect.
19	`fps`	m/m	0.00001 to `slp`	Lateral slope of the floodplain away from the channel (rise over run). Values above `slp` are lowered to `slp`; values of 10^{-6} or less are set to 10^{-5}.
20	`fpn`	-	0.025 to 0.16	Manning's n of the floodplain. Chow (1959) gives 0.025 to 0.05 for pasture and cultivated land, 0.035 to 0.16 for brush and 0.08 to 0.2 for trees.
21	`n_conc`	mg/kg	-	Nitrogen content of eroded bank sediment, added to the flow as organic N.

#	Column	Units	Typical range	Description
22	p_conc	mg/kg	-	Phosphorus content of eroded bank sediment.
23	p_bio	fraction	0 to 1	Fraction of the bank phosphorus that enters the flow as soluble P. The rest enters as sediment-bound P.

There are no built-in values for this file. A channel whose cha_hyd name is not found uses row 1. Without hyd-sed-lte.cha the chandeg channels have no parameters, so the file is required whenever chandeg.con lists channels.

sed_nut.cha

An optional table with extra coefficients for the chandeg sediment and nutrient calculations. It is read only when a file named sed_nut.cha is present. Rows are matched by the same cha_hyd name as hyd-sed-lte.cha. Two header lines are skipped, then 13 values per line:

#	Column	Units	Built-in value	Description
1	name	-	-	Name, matched against cha_hyd in channel-lte.cha.
2	order	-	-	Text field, read and not used.
3	pk_rto	ratio	1.0	Multiplier on the peak-to-mean daily flow ratio. See Peak flow.
4	fp_inun_days	days	5.0	Days the floodplain stays inundated after a flood. Replaced by 1.0 at the start of every day's sediment calculation, so neither the file value nor the built-in value has an effect.
5	n_setl	ratio	0.5	Ratio of N settling to sediment settling. Not used.
6	p_setl	ratio	0.5	Ratio of P settling to sediment settling. Not used.
7	n_sol_part	1/h	0.01	Rate of nitrate conversion to organic N, used only with the simplified transformations. See Nutrients.
8	p_sol_part	1/h	0.01	Rate of soluble P conversion to sediment-bound P, used only with the simplified transformations.
9	n_dep_enr	ratio	0.5	Enrichment ratio for organic N deposited with sediment on the floodplain and in the channel.
10	p_dep_enr	ratio	0.5	Enrichment ratio for sediment-bound P deposited with sediment.
11	arc_len_fr	fraction	1.2	Fraction of the meander arc length where banks erode. Read but not used in the erosion mass in 62.0.1.
12	bed_exp	-	1.5	Exponent of the bed erosion equation. Replaced by 1.0 at the start of every day's sediment calculation.
13	wash_bed_fr	fraction	0.1	Fraction of bank erosion deposited in the channel bed. See Channel deposition.

Without sed_nut.cha, every channel uses the built-in values. With the file present, a channel whose name is not found uses row 1.

nutrients.cha

Free format. Two header lines are skipped, then one parameter set per line with 39 values in this order. The reader does not match columns by header name. Every value of zero or less in the columns that have a value in the "Replacement" column is replaced by that value.

All rates are for 20 °C and are adjusted to the water temperature in the calculation. The water temperature used here is $5 + 0.75\,\bar{T}$, where $\bar{T}$ is the day's mean air temperature at the channel's weather station, with a minimum of 0.1 °C. The stream temperature from the heat balance (see `temperature.cha`) sets only the outflow temperature.

#	Column	Units	Replacement	Typical range	Description
1	`name`	-	-	-	Name, matched against `cha_nut` in `channel-lte.cha`.
2	`plt_n`	ppm	-	0 to 100	Organic N concentration. Not used.
3	`ptl_p`	ppm	-	0 to 100	Organic P concentration, called `plt_p` in earlier documentation. Not used.
4	`alg_stl`	m/day	1.0	0.15 to 1.82	Algal settling rate.
5	`ben_disp`	mg P/m²/day	0.05	0.001 to 0.1	Benthic source rate for dissolved P.
6	`ben_nh3n`	mg N/m²/day	0.5	0 to 1	Benthic source rate for ammonium N.
7	`ptln_stl`	1/day	0.05	0.001 to 0.1	Organic N settling rate.
8	`ptlp_stl`	1/day	0.05	0.001 to 0.1	Organic P settling rate.
9	`cst_stl`	1/day	2.5	0.01 to 10	Settling rate of an arbitrary non-conservative constituent. Not used.
10	`ben_cst`	mg/m²/day	2.5	0.01 to 10	Benthic source rate of an arbitrary non-conservative constituent. Not used.
11	`cbn_bod_co`	1/day	1.71	0.02 to 3.4	CBOD deoxygenation rate.
12	`air_rt`	1/day	1.0	0 to 100	Reaeration rate (Fickian diffusion).
13	`cbn_bod_stl`	1/day	none	-0.36 to 0.36	Rate of CBOD loss by settling. A value of zero or less is kept.
14	`ben_bod`	mg O_2/m²/day	2.0	0 to 100	Sediment oxygen demand.
15	`bact_die`	1/day	2.0	0.05 to 4	Coliform die-off rate. Not used; pathogen die-off comes from the pathogen database.
16	`cst_decay`	1/day	1.71	0 to 10	Decay rate of an arbitrary non-conservative constituent. Not used.
17	`nh3n_no2n`	1/day	0.55	0.1 to 1	Rate of biological oxidation of NH_3 to NO_2.
18	`no2n_no3n`	1/day	1.1	0.2 to 2	Rate of biological oxidation of NO_2 to NO_3.
19	`ptln_nh3n`	1/day	0.21	0.2 to 0.4	Rate of hydrolysis of organic N to ammonium.
20	`ptlp_solp`	1/day	0.35	0.01 to 0.7	Rate of mineralisation of organic P to dissolved P.
21	`q2e_lt`	-	2	1 to 4	QUAL2E light averaging option. Not used.
22	`q2e_alg`	-	2	1 to 3	Algal growth limitation: `1` multiplicative ($\mu_{max} f_L f_N f_P$), `2` limiting nutrient ($\mu_{max} f_L \min(f_N, f_P)$), `3` harmonic mean of f_N and f_P. Integer.
23	`chla_alg`	µg chl-a/mg algae	50	10 to 100	Ratio of chlorophyll-a to algal biomass.
24	`alg_n`	mg N/mg algae	0.08	0.07 to 0.09	Fraction of algal biomass that is N.
25	`alg_p`	mg P/mg algae	0.015	0.01 to 0.02	Fraction of algal biomass that is P.

#	Column	Units	Replacement	Typical range	Description
26	`alg_o2_prod`	mg O_2/mg algae	1.6	1.4 to 1.8	O_2 production per unit of algal photosynthesis.
27	`alg_o2_resp`	mg O_2/mg algae	2.0	1.6 to 2.3	O_2 uptake per unit of algal respiration.
28	`o2_nh3n`	mg O_2/mg N	3.5	3 to 4	O_2 uptake per unit of NH_3 oxidised.
29	`o2_no2n`	mg O_2/mg N	1.07	1 to 1.4	O_2 uptake per unit of NO_2 oxidised.
30	`alg_grow`	1/day	2.0	1 to 3	Maximum specific algal growth rate.
31	`alg_resp`	1/day	2.5	0.05 to 5	Algal respiration rate.
32	`slr_act`	fraction	0.3	0 to 1	Fraction of solar radiation that is photosynthetically active.
33	`lt_co`	kJ/m²/min	0.75	-	Half-saturation coefficient for light. Multiplied by 0.06 on reading, which converts kJ/m²/min to MJ/m²/h.
34	`const_n`	mg N/L	0.02	0.01 to 0.3	Michaelis-Menten half-saturation constant for N.
35	`const_p`	mg P/L	0.025	0.001 to 0.05	Michaelis-Menten half-saturation constant for P.
36	`lt_nonalg`	1/m	1.0	0 to 10	Non-algal part of the light extinction coefficient.
37	`alg_shd_l`	1/(m · µg chl-a/L)	0.03	0.006 to 0.065	Linear algal self-shading coefficient.
38	`alg_shd_nl`	(1/m)(µg chl-a/L)^-2/3	0.054	0 to 1	Non-linear algal self-shading coefficient.
39	`nh3_pref`	-	0.5	0 to 1	Algal preference for ammonium over nitrate.

Units of the half-saturation coefficient for light differ between sources: the old GitBook pages give `lt_co` in MJ/m²/h with a range of 0.223 to 1.135, while the 62.0.1 reader treats the value as kJ/m²/min and multiplies it by 0.06. A file value of 0.75 is used as 0.045 MJ/m²/h.

In sub-daily runs the reader divides the rates in columns 4 to 8, 11 to 14, 17 to 20, 30 and 31 by the number of time steps per day, while the water quality calculation still runs once per day. Sub-daily runs therefore use smaller effective rates than daily runs with the same file.

When `nutrients.cha` is missing, the chandeg channels have no water quality parameters, so keep the file in any model with channels.

Channel geometry and rating curve

At the start of the run each channel gets a trapezoidal cross-section and a rating curve at four depths: 0.1 and 1.0 times the bankfull depth `dp`, and 1.2 and 2.0 times `dp` with water on the floodplain. Flow, velocity, area, volume and travel time at any other flow are interpolated between these points.

With top width W (`wd`), bankfull depth D (`dp`), side slope z (`side_slp`), length L (`len`, km), slope S (`slp`) and roughness n (`mann`), the bottom width is

$$b = W - 2\,D\,z$$

If this is zero or negative, b is set to $W/2$ and z is reset to $W/(4D)$. For a flow depth d up to bankfull,

$$P = b + 2\,d\,\sqrt{1 + 1/z^2}, \qquad A = b\,d + d/z, \qquad R = A/P$$

$$v = \frac{R^{2/3}\,S^{1/2}}{n + 0.001}, \qquad Q = A\,v, \qquad TT = \frac{L}{3.6\,v}$$

where P is the wetted perimeter (m), A the flow area (m²), v the velocity (m/s), Q the flow (m³/s) and TT the travel time (h). The channel volume is $1000\,A\,L$ (m³).

The bottom width treats `side_slp` as horizontal distance per unit rise, while the perimeter and area treat it as rise per unit horizontal distance. The side term of the area, d/z, is the area of two side triangles with slope z (rise over run) only at $d = 1$ m. These are the expressions SWAT+ 62.0.1 uses.

Above bankfull, with h the depth above the bank (0.2 D and 1.0 D) and s_{fp} the floodplain slope `fps`,

$$P = P_{bf} + 2\,h\,\sqrt{1 + 1/s_{fp}^2}, \qquad A = A_{bf} + b\,h + h^2/s_{fp}, \qquad W_{fp} = W + 2\,h/s_{fp}$$

The floodplain volume is $1000\,W_{fp}\,h\,L$, and flow and velocity use `fpn` in place of `mann`, with the channel slope `slp`.

A smaller `fps` gives a wider, flatter floodplain that stores more water at a given depth.

Flow routing

Each day the inflow from upstream objects is first increased by any water allocation transfer into the channel, aquifer flow from `aqu_cha.lin`, gwflow exchange when gwflow is active, and, on days with inflow, precipitation on the channel surface ($P\,L\,W$ m³, with P in mm, L in km and W in m). Water is then routed with one of two methods, chosen by `rte_cha` in `codes.bsn`.

Variable storage method

With `rte_cha` = 0, the outflow in each time step is a fraction of the stored water:

$$O = c\,V_{stor}, \qquad c = \min\left(1,\ \frac{2\,s_c\,\Delta t}{2\,TT + \Delta t}\right)$$

where s_c is `scoef` in `parameters.bsn`, Δt the time step (h) and TT the travel time (h) at the previous step's outflow rate.

Muskingum method

With `rte_cha` = 1, outflow follows the Muskingum equation

$$O_2 = C_1 I_2 + C_2 I_1 + C_3 O_1$$

The storage constant is a weighted mean of the travel times at bankfull and 0.1 bankfull depth, each computed with the kinematic wave celerity $5v/3$:

$$K = \frac{\text{msk_co1}\,K_{bf} + \text{msk_co2}\,K_{0.1}}{\text{msk_co1} + \text{msk_co2}}$$

With X = `msk_x` (all three in `parameters.bsn`), the coefficients are

$$C_1 = \frac{\Delta t - 2KX}{2K(1-X) + \Delta t}, \quad C_2 = \frac{\Delta t + 2KX}{2K(1-X) + \Delta t}, \quad C_3 = \frac{2K(1-X) - \Delta t}{2K(1-X) + \Delta t}$$

C_1 is kept at zero or above, and the three are scaled to sum to 1. When $\Delta t > 2K(1-X)$ the coefficients are computed for a shorter sub-step that meets this limit. Sub-steps are used only in sub-daily runs; a daily run routes once per day with the sub-step coefficients.

Stored water above the bankfull channel volume is counted as floodplain storage.

Water losses

After routing, transmission losses and evaporation are taken from the water left in the channel:

$$TL = 24\,k\,L\,W \quad (\text{m}^3/\text{day}), \qquad E = \text{evrch}\ PET\ L\,W$$

where k is `k` (mm/h), L is in km, W in m, PET is the potential evapotranspiration of the weather station (mm), and `evrch` is the reach evaporation factor (`evap_adj` in `parameters.bsn`). Both use the bankfull top width over the full channel length, and neither can exceed the stored volume. With gwflow active, gwflow also calculates an exchange with the channel; the transmission loss above is still taken from the channel, but the channel output reports it as zero seepage.

Sediment

Sediment calculations run once per day for channels with inflow. In 62.0.1 four inputs are replaced at the start of each day's calculation: the bankfull flow multiplier by 1.5, the floodplain inundation days by 1.0, the bed `d50` by 1.0 mm and the bed erosion exponent by 1.0. The file values of `bed_load`, `d50`, `fp_inun_days` and `bed_exp`, and any calibration changes to them, therefore have no effect.

Peak flow

The day's peak flow is

$$q_{pk} = \text{pk_rto}\,(1 + 2.66\,A_{km^2}^{-0.3})\,\frac{Q_{day}}{86400}$$

(Fuller's equation), where A_{km^2} is the `area` of the channel in `chandeg.con` converted to km² and Q_{day} is the day's inflow (m³). The peak sets the velocity v from the rating curve, which is used for erosion and the travel time.

Floodplain deposition

When q_{pk} exceeds 1.5 times the bankfull flow, the floodplain traps a fraction of the incoming sediment (Dynamic SedNet, 2017):

$$TE = \min\!\left(1,\ N_{inun}\,\frac{q_{fp}}{\bar{q}}\,\left(1 - e^{-0.0001\,A_{fp}/q_{fp}}\right)\right)$$

with $N_{inun} = 1$ day, q_{fp} the rating curve flow at the peak, $\bar{q}$ the mean daily flow (m³/s) and $A_{fp} = 5\,W\,L$ the floodplain area (m², L in m). Organic N and sediment-bound P are deposited with TE times `n_dep_enr` and `p_dep_enr`. Overbank water then fills wetlands of HRUs linked to the channel's floodplain up to their emergency volume.

Bank erosion

Banks erode when the reach velocity exceeds a critical velocity. The cohesion from the clay content C (`clay`, %) is

$$c = \max(1000,\ -87.1 + 42.82\,C - 0.261\,C^2 + 0.029\,C^3)$$

and the critical velocity is

$$v_{cr} = \text{eq_slp}\ \log_{10}(2200\,D)\,\sqrt{0.0004\,(f_{bd} + 0.021\,c + 4500\,\text{cov}\,e^{-5D})}$$

where $f_{bd} = \max(0.001,\ 39.24\,\rho_b - 1000)$ with ρ_b the bulk density `dry_bd` (t/m³). For bulk densities below 25.5 t/m³, which covers all real sediments, f_{bd} stays at 0.001, so `dry_bd` does not change v_{cr}.

The velocity at the bends uses the radius of curvature from the sinuosity σ (column `wd_rto`):

$$r_c = \frac{12\,W\,\sigma^{1.5}}{13\,\sqrt{\sigma - 1}}, \qquad v_{rch} = 0.33\,v\,(1 + 1/r_c) + 0.66\,v$$

If $v_{rch} > v_{cr}$, the bank retreat for the day (m) is

$$e_{bank} = \left(\frac{1}{v_c} + \frac{1}{0.0002\,W}\right)^{-1}, \qquad v_c = \frac{0.0001\,A_{km^2}^{-0.0858}\,W}{1 + e^{-4\,(v_{rch}/v_{cr} - 1)}}$$

and the eroded mass is $1000\,e_{bank}\,D\,(0.25\,L)\,\rho_b$ tonnes, with L in km. The bank sediment brings organic N at `n_conc` and P at `p_conc`; the fraction `p_bio` of that P is soluble.

Channel deposition

A fraction `wash_bed_fr` of the bank erosion is deposited in the channel bed and removed from the flow, with organic N and sediment-bound P scaled by `n_dep_enr` and `p_dep_enr`.

Bed erosion

For channels steeper than 10^{-6}, bed erosion occurs when $v > 0.293\sqrt{d_{50}}$, which is 0.293 m/s with $d_{50} = 1$ mm:

$$e_{bed} = 0.0001\left(\frac{v}{0.293}\right)\ \text{m}, \qquad M_{bed} = 1000\, e_{bed}\, W\, L\, \rho_b\ \text{t}$$

In 62.0.1, e_{bed} and M_{bed} appear in the channel morphology and sediment budget outputs, but the eroded bed sediment is not added to the flow.

Nutrients

For channels with inflow, the day's water quality follows the QUAL2E kinetics with the rates of `nutrients.cha`, applied over the travel time (at most one day). This covers algae and chlorophyll-a, CBOD, dissolved oxygen, the nitrogen chain from organic N through ammonium and nitrite to nitrate, and organic and soluble P. See Channels in the model theory.

When column 24 of `codes.bsn` (headed `i_fpwet` by the Editor) is 1, a simplified transformation then replaces the organic N, nitrate, sediment-bound P and soluble P of the outflow:

$$\Delta N_{org} = \min\left(NO_3,\ \left(1 - e^{-k_N\, TT}\right) N_{org}\right)$$

with $k_N =$ `n_sol_part` and TT the travel time (h). ΔN_{org} is added to organic N and taken from nitrate. Phosphorus follows the same form with `p_sol_part`, moving soluble P to sediment-bound P.

Pesticides

In the channel, pesticide sorption uses $K_d = K_{oc}\,\text{carbon}/100$, with K_{oc} from `pesticide.pes` and `carbon` from `hyd-sed-lte.cha`. The mixing velocity between water and bed uses the bed porosity $1 - \rho_b/2.65$. Pesticides and pathogens are routed only on days with inflow.

Calibration and decision tables

Parameters can be changed during calibration through `calibration.cal` (see parameter change files), using these names:

- `hyd-sed-lte.cha`: `chw` (`wd`), `chd` (`dp`), `chs` (`slp`), `chl` (`len`), `chn` (`mann`), `chk` (`k`), `bank_exp` (`erod_fact`), `cov` (`cov_fact`), `vcr_coef` (`eq_slp`), `d50`, `ch_clay` (`clay`), `carbon`, `ch_bd` (`dry_bd`), `chss` (`side_slp`), `bankfull_flo` (`bed_load`), `fps`, `fpn`, `ch_n_conc`, `ch_p_conc`, `ch_p_bio`.
- `sed_nut.cha`: `pk_rto`, `fp_inun_days`, `n_setl`, `p_setl`, `n_sol_part`, `p_sol_part`, `n_dep_enr`, `p_dep_enr`, `arc_len_fr`, `bed_exp`, `wash_bed_fr`.
- `nutrients.cha`: `rs1` to `rs7`, `rk1` to `rk4`, `rk6`, `bc1` to `bc4`, `mumax`. The parameter name `rk5` changes `ben_disp` (the same column as `rs2`); no calibration name changes `bact_die`.

Calibration changes are applied after the rating curve and Muskingum coefficients are set, and these are not recalculated. Changes to `chw`, `chd`, `chs`, `chn`, `chss`, `fps` and `fpn` therefore do not change the rating curve or the routing coefficients; they change only the other calculations that use the values directly (transmission losses, evaporation, erosion). Changes to `bankfull_flo`, `fp_inun_days`, `d50` and `bed_exp` are overwritten each day (see Sediment). The `nutrients.cha` changes are applied to the row whose position equals the channel number.

The decision table action `chan_change` sets a channel's cover factor to the action's first constant and its stream order to the second. The condition `ch_order` tests a channel's stream order. See decision tables.

Example

`refdata/Osu_1hru` in the SWAT+ repository, first rows of each file:

```
channel-lte.cha: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                        cha_ini           cha_hyd           cha_sed           cha_nut
       1  cha01                      initcha1          hydcha01              null           nutcha1
       2  cha03                      initcha1          hydcha03              null           nutcha1

hyd-sed-lte.cha: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name        order       wd       dp      slp      len     mann        k  erod_fact  cov_fact   wd_rto   eq
hydcha01        4 27.17726  0.99164  0.00146  3.42760  0.05000  5.00000    0.01000   0.00500 27.40638  0.0
```

(columns narrowed for display)

Channel `cha01` is a fourth-order channel 27.2 m wide, 0.99 m deep and 3.43 km long, with Manning's n of 0.05 and bed conductivity of 5 mm/h. Some values in this file are read differently from what the Editor header suggests:

- `wd_rto` = 27.4 is the width-to-depth ratio, but SWAT+ reads it as a sinuosity of 27.4. The bends then have a large radius of curvature, and the bend velocity is close to the mean velocity.
- `eq_slp` = 0.001 multiplies the critical velocity for bank erosion by 0.001, so bank erosion occurs on almost every day with flow.
- `d50` = 12 and `bed_load` = 0.5 are replaced by 1.0 mm and 1.5 during the run.
- `fps` = 0.00001 gives a very flat, wide floodplain.

`nutrients.cha` in the same dataset has one set, `nutcha1`, with `air_rt` = 50 and `cbn_bod_stl` = 0.36. The other values are the replacement values in the table above, or zero for the two unused concentrations.

Related files

- `chandeg.con`: lists the channels and points to `channel-lte.cha` rows in its `lcha` column.
- `initial.cha`: initial water, sediment and constituents in each channel, through `om_water.ini`, `pest_water.ini`, `salt_channel.ini` and `cs_channel.ini`.
- `temperature.cha`: stream temperature parameters.
- `codes.bsn` (`rte_cha`, column 24) and `parameters.bsn` (`msk_co1`, `msk_co2`, `msk_x`, `evap_adj`, `scoef`): routing switches and coefficients.
- `file.cio`: the `channel` line names the files.
- Outputs: channel, sediment and nutrient files (`channel_sd`, `channel_sdmorph`, `sd_chanbud`, `ch_sedbud`, `chanbud_order`).
- Channels and erosion and sediment in the model theory.

Sources

- File name slots: `type input_cha` in `input_file_module.f90`, read in `readcio_read.f90`; read order in `proc_cha.f90`
- Readers: `sd_channel_read.f90` (`channel-lte.cha`), `sd_hydsed_read.f90` (`hyd-sed-lte.cha`, `sed_nut.cha`), `ch_read_nut.f90` (`nutrients.cha`)
- Type definitions: `swatdeg_datafiles`, `swatdeg_hydsed_data`, `swatdeg_sednut_data` and `swatdeg_channel_dynamic` in `sd_channel_module.f90`; `channel_nut_data` in `channel_data_module.f90`
- Initialisation, limits, rating curve and Muskingum coefficients: `sd_hydsed_init.f90`, `sd_rating_curve.f90`, `qman.f90`
- Daily channel calculation: `sd_channel_control3.f90`, called from `command.f90`; sediment in `sd_channel_sediment3.f90`; routing and losses in `ch_rtmusk.f90`; water quality in `ch_watqual4.f90`; pesticides in `ch_rtpest.f90`
- `chandeg.con` pointer column: `hyd_read_connect.f90`
- Calibration: `cal_parm_select.f90`; decision tables: `actions.f90`, `conditions.f90`

`channel-lte.cha`:

Column	Fortran variable
`id`	index of `sd_dat`
`name`	`sd_dat%name`
`cha_ini`	`sd_dat%initc` -> `sd_dat%init`
`cha_hyd`	`sd_dat%hydc` -> `sd_dat%hyd`, `sd_dat%sednut`

Column	Fortran variable
cha_sed	sd_dat%sedc (unused)
cha_nut	sd_dat%nutc -> sd_dat%nut

hyd-sed-lte.cha (copied to sd_ch in sd_hydsed_init):

Column	Fortran variable
order	sd_chd%order
wd	sd_chd%chw
dp	sd_chd%chd
slp	sd_chd%chs
len	sd_chd%chl
mann	sd_chd%chn
k	sd_chd%chk
erod_fact	sd_chd%bank_exp
cov_fact	sd_chd%cov
wd_rto	sd_chd%sinu
eq_slp	sd_chd%vcr_coef
d50	sd_chd%d50
clay	sd_chd%ch_clay
carbon	sd_chd%carbon
dry_bd	sd_chd%ch_bd
side_slp	sd_chd%chss
bed_load	sd_chd%bankfull_flo
fps, fpn, n_conc, p_conc, p_bio	sd_chd% same names

sed_nut.cha columns map to sd_chd1% components of the same names.

nutrients.cha:

Columns	Fortran variables (ch_nut%)
plt_n, ptl_p	onco, opco
alg_stl to ben_cst	rs1 to rs7
cbn_bod_co, air_rt, cbn_bod_stl, ben_bod, bact_die, cst_decay	rk1 to rk6
nh3n_no2n to ptlp_solp	bc1 to bc4
q2e_lt, q2e_alg	lao, igropt
chla_alg to o2_no2n	ai0 to ai6
alg_grow, alg_resp, slr_act	mumax, rhoq, tfact
lt_co, const_n, const_p	k_l, k_n, k_p
lt_nonalg, alg_shd_l, alg_shd_nl, nh3_pref	lambda0, lambda1, lambda2, p_n